cosmic microwave background radiation: Remnant electromagnetic radiation (photons) from the early universe, which permeates space.

cosmic horizon, horizon: Locations in space beyond which light has not had time to reach us, since the beginning of the universe.

cosmological constant: A hypothetical energy and pressure, uniformly filling space; origin and composition unknown.

cosmology: Study of origin and evolution of the universe. critical density: Amount of mass/energy density required for space to be flat; about 10^{-23} grams per cubic meter.

D-branes, Dirichlet-*p*-branes: A *p*-brane that is "sticky"; a *p*-brane to which open string endpoints are attached.

dark energy: A hypothetical energy and pressure, uniformly filling space; more general notion than a cosmological constant as its energy/pressure can vary with time.

dark matter: Matter suffused through space, exerting gravity but not emitting light.

electromagnetic field: The field which exerts the electromagnetic force.

electromagnetic force: One of nature's four forces; acts on particles that have electric charge.

electron field: The field for which the electron particle is the smallest bundle or constituent.

electroweak theory: The theory unifying the electromagnetic and the weak nuclear forces into the electroweak force.

electroweak Higgs field: Field that acquires a nonzero value in cold, empty space; gives rise to masses for fundamental particles.

energy bowl: See *potential energy bowl.*

entropy: A measure of the disorder of a physical system; the number of rearrangements of a system's fundamental constituents that leave its gross, overall appearance unchanged.

entanglement, quantum entanglement: Quantum phenomenon in which spatially distant particles have correlated properties.

event horizon: Imaginary sphere surrounding a black hole delineating the points of no return; anything crossing the event horizon cannot escape the black hole's gravity. field: A "mist" or "essence" permeating space; can convey a force or describe the presence/motion of particles. Mathematically, involves a number or collection of numbers at each point in space, signifying the field's value.

flat space: Possible shape of the spatial universe having no curvature.

flatness problem: Challenge for cosmological theories to explain observed flatness of space.

general relativity: Einstein's theory of gravity; invokes curvature of space and time.

gluons: Messenger particles of the strong nuclear force.